

Information About the Performed Analysis

You chose to analyze the following complexity measures:

1. C_{len} (n traces)
2. $C_{\text{DT-}\#}$ (n variants)
3. $C_{\text{DT-}\%}$ (ratio variants per number of traces)
4. C_{mag} (n events)
5. $C_{\text{TL-min}}$ (trace len min)
6. $C_{\text{TL-max}}$ (trace len max)
7. $C_{\text{TL-mean}}$ (trace len mean)
8. $C_{\text{TL-med}}$ (trace len median)
9. $C_{\text{TL-mode}}$ (trace len mode)
10. $C_{\text{TL-dev}}$ (trace len std)
11. $C_{\text{TL-var}}$ (trace len variance)
12. $C_{\text{TL-Q1}}$ (trace len q1)
13. $C_{\text{TL-Q3}}$ (trace len q3)
14. $C_{\text{TL-IQR}}$ (trace len iqr)
15. $C_{\text{TL-g-mean}}$ (trace len geometric mean)
16. $C_{\text{TL-g-dev}}$ (trace len geometric std)
17. $C_{\text{TL-h-mean}}$ (trace len harmonic mean)
18. $C_{\text{TL-skew}}$ (trace len skewness)
19. $C_{\text{TL-kurt}}$ (trace len kurtosis)
20. $C_{\text{TL-cvar}}$ (trace len coefficient variation)
21. $C_{\text{TL-ent}}$ (trace len entropy)
22. $C_{\text{TL-hist1}}$ (trace len hist1)
23. $C_{\text{TL-hist2}}$ (trace len hist2)
24. $C_{\text{TL-hist3}}$ (trace len hist3)
25. $C_{\text{TL-hist4}}$ (trace len hist4)
26. $C_{\text{TL-hist5}}$ (trace len hist5)
27. $C_{\text{TL-hist6}}$ (trace len hist6)
28. $C_{\text{TL-hist7}}$ (trace len hist7)
29. $C_{\text{TL-hist8}}$ (trace len hist8)
30. $C_{\text{TL-hist9}}$ (trace len hist9)
31. $C_{\text{TL-hist10}}$ (trace len hist10)
32. $C_{\text{TL-hist-skew}}$ (trace len skewness hist)
33. $C_{\text{TL-hist-kurt}}$ (trace len kurtosis hist)
34. $C_{\text{FREQ-T-max-r}}$ (ratio most common variant)
35. $C_{\text{FREQ-T-max-1\%}}$ (ratio top 1 variants)
36. $C_{\text{FREQ-T-max-5\%}}$ (ratio top 5 variants)
37. $C_{\text{FREQ-T-max-10\%}}$ (ratio top 10 variants)
38. $C_{\text{FREQ-T-max-20\%}}$ (ratio top 20 variants)
39. $C_{\text{FREQ-T-max-50\%}}$ (ratio top 50 variants)
40. $C_{\text{FREQ-T-max-75\%}}$ (ratio top 75 variants)
41. $C_{\text{FREQ-T-mean}}$ (mean variant occurrence)
42. $C_{\text{FREQ-T-dev}}$ (std variant occurrence)
43. $C_{\text{FREQ-T-skew}}$ (skewness variant occurrence)
44. $C_{\text{FREQ-T-kurt}}$ (kurtosis variant occurrence)
45. $C_{\text{DT-cov}}$ (coverage variants)
46. $C_{\text{DT-cov-}\%}$ (rel coverage variants)
47. $C_{\text{DT-het}}$ (heterogeneity rate variants)
48. $C_{\text{DT-sim}}$ (similarity rate variants)
49. C_{var} (n unique activities)
50. $C_{\text{FREQ-A-min}}$ (activities min)
51. $C_{\text{FREQ-A-max}}$ (activities max)
52. $C_{\text{FREQ-A-mean}}$ (activities mean)
53. $C_{\text{FREQ-A-med}}$ (activities median)
54. $C_{\text{FREQ-A-dev}}$ (activities std)
55. $C_{\text{FREQ-A-var}}$ (activities variance)

56. $C_{\text{FREQ-A-Q1}}$ (activities q1)
57. $C_{\text{FREQ-A-Q3}}$ (activities q3)
58. $C_{\text{FREQ-A-IQR}}$ (activities iqr)
59. $C_{\text{FREQ-A-skew}}$ (activities skewness)
60. $C_{\text{FREQ-A-kurt}}$ (activities kurtosis)
61. $C_{\text{A-start-}\#}$ (n unique start activities)
62. $C_{\text{A-start-min}}$ (start activities min)
63. $C_{\text{A-start-max}}$ (start activities max)
64. $C_{\text{A-start-mean}}$ (start activities mean)
65. $C_{\text{A-start-med}}$ (start activities median)
66. $C_{\text{A-start-dev}}$ (start activities std)
67. $C_{\text{A-start-var}}$ (start activities variance)
68. $C_{\text{A-start-Q1}}$ (start activities q1)
69. $C_{\text{A-start-Q3}}$ (start activities q3)
70. $C_{\text{A-start-IQR}}$ (start activities iqr)
71. $C_{\text{A-start-skew}}$ (start activities skewness)
72. $C_{\text{A-start-kurt}}$ (start activities kurtosis)
73. $C_{\text{A-start-rel}}$ (rel unique start activities)
74. $C_{\text{A-end-}\#}$ (n unique end activities)
75. $C_{\text{A-end-min}}$ (end activities min)
76. $C_{\text{A-end-max}}$ (end activities max)
77. $C_{\text{A-end-mean}}$ (end activities mean)
78. $C_{\text{A-end-med}}$ (end activities median)
79. $C_{\text{A-end-dev}}$ (end activities std)
80. $C_{\text{A-end-var}}$ (end activities variance)
81. $C_{\text{A-end-Q1}}$ (end activities q1)
82. $C_{\text{A-end-Q3}}$ (end activities q3)
83. $C_{\text{A-end-IQR}}$ (end activities iqr)
84. $C_{\text{A-end-skew}}$ (end activities skewness)
85. $C_{\text{A-end-kurt}}$ (end activities kurtosis)
86. $C_{\text{A-end-rel}}$ (rel unique end activities)
87. $C_{\text{ENT-trace}}$ (eventropy trace)
88. $C_{\text{ENT-pref}}$ (eventropy prefix)
89. $C_{\text{ENT-pref-F}}$ (eventropy prefix flattened)
90. $C_{\text{ENT-g-block}}$ (eventropy global block)
91. $C_{\text{ENT-g-block-F}}$ (eventropy global block flattened)
92. $C_{\text{ENT-LZ}}$ (eventropy lempel ziv)
93. $C_{\text{ENT-LZ-F}}$ (eventropy lempel ziv flattened)
94. $C_{\text{ENT-k-block-d1}}$ (eventropy k block diff 1)
95. $C_{\text{ENT-k-block-d3}}$ (eventropy k block diff 3)
96. $C_{\text{ENT-k-block-d5}}$ (eventropy k block diff 5)
97. $C_{\text{ENT-k-block-r1}}$ (eventropy k block ratio 1)
98. $C_{\text{ENT-k-block-r3}}$ (eventropy k block ratio 3)
99. $C_{\text{ENT-k-block-r5}}$ (eventropy k block ratio 5)
100. $C_{\text{ENT-knn-3}}$ (eventropy knn 3)
101. $C_{\text{ENT-knn-5}}$ (eventropy knn 5)
102. $C_{\text{ENT-knn-7}}$ (eventropy knn 7)
103. $C_{\text{epa-var}}$ (epa variant entropy)
104. $C_{\text{epa-n-var}}$ (epa normalized variant entropy)
105. $C_{\text{epa-seq}}$ (epa sequence entropy)
106. $C_{\text{epa-n-seq}}$ (epa normalized sequence entropy)
107. $C_{\text{epa-seq-L}}$ (epa sequence entropy linear forgetting)
108. $C_{\text{epa-n-seq-L}}$ (epa normalized sequence entropy linear forgetting)
109. $C_{\text{epa-seq-E}}$ (epa sequence entropy exponential forgetting)
110. $C_{\text{epa-n-seq-E}}$ (epa normalized sequence entropy exponential forgetting)
111. $C_{\text{s-T-div}}$ (simple trace diversity)
112. $C_{\text{a-T-div}}$ (advanced trace diversity)
113. $C_{\text{DA-min}}$ (distinct activities min)

114. C_{DA-max} (distinct activities max)
115. C_{DA-avg} (distinct activities mean)
116. C_{DA-dev} (distinct activities std)
117. $C_{DA-dens}$ (event density)
118. $C_{DA-ovrlp}$ (distinct activities non overlap)
119. $C_{comp-fact}$ (complexity factor)
120. $C_{loop-#}$ (n traces with loop)
121. $C_{loop-rel-#}$ (avg traces with loop)
122. $C_{loop-avg-#}$ (avg loops per trace)
123. $C_{loop-max-#}$ (max loops per trace)
124. $C_{loop-avg-size}$ (avg loop size per trace)
125. $C_{loop-max-size}$ (max loop size per trace)
126. $C_{rep-#}$ (n traces with repetition)
127. $C_{rep-rel-#}$ (avg traces with repetition)
128. C_{succ} (number of successions)
129. C_{ties} (number of ties)
130. C_{struct} (structure)
131. $C_{affinity}$ (average affinity)
132. C_{LZ} (lempel ziv complexity)
133. C_{dev-R} (deviation from random)
134. $C_{avg-dist}$ (average edit distance)
135. C_{DFG-V} (n nodes dfg)
136. C_{DFG-E} (n edges dfg)
137. $C_{DFG-cnc}$ (coeff of connectivity dfg)
138. $C_{DFG-adeq}$ (avg node degree dfg)
139. $C_{DFG-mdeg}$ (max node degree dfg)
140. $C_{DFG-dens}$ (density dfg)
141. $C_{DFG-struct}$ (structure dfg)
142. C_{DFG-CN} (cyclomatic number dfg)
143. $C_{DFG-sep-#}$ (n cut vertices dfg)
144. $C_{DFG-sep-r}$ (separability ratio dfg)
145. $C_{DFG-seq-r}$ (sequentiality ratio dfg)
146. $C_{DFG-cyc}$ (cyclicity dfg)

Out of these measures, the following were excluded because they returned non-numeric values:

1. $C_{TL-skew}$ (trace len skewness)
2. $C_{TL-kurt}$ (trace len kurtosis)
3. $C_{FREQ-T-skew}$ (skewness variant occurrence)
4. $C_{FREQ-T-kurt}$ (kurtosis variant occurrence)
5. $C_{A-start-skew}$ (start activities skewness)
6. $C_{A-start-kurt}$ (start activities kurtosis)
7. $C_{A-end-skew}$ (end activities skewness)
8. $C_{A-end-kurt}$ (end activities kurtosis)

Furthermore, the following measures were excluded because their variance is 0 in the analyzed data:

1. C_{len} (n traces)
2. C_{DT-cov} (coverage variants)

Additionally, the following measures were excluded because they are correlated with other measures:

1. $C_{DT- \%}$ (ratio variants per number of traces)
2. $C_{TL-mean}$ (trace len mean)
3. $C_{DT-cov- \%}$ (rel coverage variants)
4. C_{DFG-V} (n nodes dfg)
5. $C_{ENT-k-block-r1}$ (eventropy k block ratio 1)
6. $C_{loop-rel-#}$ (avg traces with loop)
7. $C_{rep-rel-#}$ (avg traces with repetition)
8. $C_{DFG-adeq}$ (avg node degree dfg)

This leaves us with a total of 128 log measures to analyze.

Validity of the Factor Analysis

Your sample size in this analysis was 2112. It is recommended to have a sample size of at least $5 \cdot 146 = 730$. Bartlett's test of sphericity returned a χ^2 value of 1321643.419324327 and a p -value of 0.0. This means, the variables are probably correlated and we can perform a factor analysis.

The measure of sampling adequacy (MSA) returned the following results:

- $C_{DT-\#}$: 0.9521
- C_{mag} : 0.9344
- C_{TL-min} : 0.9632
- C_{TL-max} : 0.9721
- C_{TL-med} : 0.9886
- $C_{TL-mode}$: 0.9907
- C_{TL-dev} : 0.8897
- C_{TL-var} : 0.9047
- C_{TL-Q1} : 0.9715
- C_{TL-Q3} : 0.9674
- C_{TL-IQR} : 0.9543
- $C_{TL-g-mean}$: 0.9318
- $C_{TL-g-dev}$: 0.8566
- $C_{TL-h-mean}$: 0.9315
- $C_{TL-cvar}$: 0.6882
- C_{TL-ent} : 0.7195
- $C_{TL-hist1}$: 0.6888
- $C_{TL-hist2}$: 0.891
- $C_{TL-hist3}$: 0.8824
- $C_{TL-hist4}$: 0.654
- $C_{TL-hist5}$: 0.7404
- $C_{TL-hist6}$: 0.6117
- $C_{TL-hist7}$: 0.7078
- $C_{TL-hist8}$: 0.6925
- $C_{TL-hist9}$: 0.856
- $C_{TL-hist10}$: 0.6409
- $C_{TL-hist-skew}$: 0.8819
- $C_{TL-hist-kurt}$: 0.8898
- $C_{FREQ-T-max-r}$: 0.8324
- $C_{FREQ-T-max-1\%}$: 0.975
- $C_{FREQ-T-max-5\%}$: 0.8917
- $C_{FREQ-T-max-10\%}$: 0.9036
- $C_{FREQ-T-max-20\%}$: 0.8457
- $C_{FREQ-T-max-50\%}$: 0.8209
- $C_{FREQ-T-max-75\%}$: 0.9571
- $C_{FREQ-T-mean}$: 0.9162
- $C_{FREQ-T-dev}$: 0.9152
- C_{DT-het} : 0.7606
- C_{DT-sim} : 0.7519
- C_{var} : 0.704
- $C_{FREQ-A-min}$: 0.8737
- $C_{FREQ-A-max}$: 0.9722
- $C_{FREQ-A-mean}$: 0.9341
- $C_{FREQ-A-med}$: 0.9722
- $C_{FREQ-A-dev}$: 0.8861
- $C_{FREQ-A-var}$: 0.9002
- $C_{FREQ-A-Q1}$: 0.9282
- $C_{FREQ-A-Q3}$: 0.9477
- $C_{FREQ-A-IQR}$: 0.941
- $C_{FREQ-A-skew}$: 0.8473
- $C_{FREQ-A-kurt}$: 0.7899
- $C_{A-start-\#}$: 0.8213
- $C_{A-start-min}$: 0.7807

- $C_{A\text{-start-max}}$: 0.7112
- $C_{A\text{-start-mean}}$: 0.8069
- $C_{A\text{-start-med}}$: 0.8244
- $C_{A\text{-start-dev}}$: 0.7942
- $C_{A\text{-start-var}}$: 0.8836
- $C_{A\text{-start-Q1}}$: 0.8877
- $C_{A\text{-start-Q3}}$: 0.8677
- $C_{A\text{-start-rel}}$: 0.7171
- $C_{A\text{-end-}}$: 0.8988
- $C_{A\text{-end-min}}$: 0.8136
- $C_{A\text{-end-max}}$: 0.9151
- $C_{A\text{-end-mean}}$: 0.8522
- $C_{A\text{-end-med}}$: 0.8705
- $C_{A\text{-end-dev}}$: 0.7412
- $C_{A\text{-end-var}}$: 0.9164
- $C_{A\text{-end-Q1}}$: 0.9268
- $C_{A\text{-end-Q3}}$: 0.8923
- $C_{A\text{-end-rel}}$: 0.7307
- $C_{\text{ENT-trace}}$: 0.9559
- $C_{\text{ENT-pref}}$: 0.9298
- $C_{\text{ENT-pref-F}}$: 0.9216
- $C_{\text{ENT-g-block}}$: 0.9202
- $C_{\text{ENT-g-block-F}}$: 0.9171
- $C_{\text{ENT-LZ}}$: 0.9351
- $C_{\text{ENT-LZ-F}}$: 0.9014
- $C_{\text{ENT-k-block-d1}}$: 0.6808
- $C_{\text{ENT-k-block-d3}}$: 0.8091
- $C_{\text{ENT-k-block-d5}}$: 0.9485
- $C_{\text{ENT-k-block-r3}}$: 0.8061
- $C_{\text{ENT-k-block-r5}}$: 0.8338
- $C_{\text{ENT-knn-3}}$: 0.9213
- $C_{\text{ENT-knn-5}}$: 0.9153
- $C_{\text{ENT-knn-7}}$: 0.9151
- $C_{\text{epa-var}}$: 0.966
- $C_{\text{epa-n-var}}$: 0.7655
- $C_{\text{epa-seq}}$: 0.95
- $C_{\text{epa-n-seq}}$: 0.7835
- $C_{\text{epa-seq-L}}$: 0.9489
- $C_{\text{epa-n-seq-L}}$: 0.7367
- $C_{\text{epa-seq-E}}$: 0.9486
- $C_{\text{epa-n-seq-E}}$: 0.7584
- $C_{s\text{-T-div}}$: 0.8646
- $C_{a\text{-T-div}}$: 0.8852
- $C_{\text{DA-max}}$: 0.9465
- $C_{\text{DA-avg}}$: 0.9261
- $C_{\text{DA-dev}}$: 0.9207
- $C_{\text{DA-dens}}$: 0.8069
- $C_{\text{DA-ovrlp}}$: 0.8892
- $C_{\text{comp-fact}}$: 0.9079
- $C_{\text{loop-}}$: 0.8283
- $C_{\text{loop-avg-}}$: 0.8291
- $C_{\text{loop-max-}}$: 0.9313
- $C_{\text{loop-avg-size}}$: 0.9071
- $C_{\text{loop-max-size}}$: 0.8588
- $C_{\text{rep-}}$: 0.9224
- C_{succ} : 0.9443
- C_{ties} : 0.9431
- C_{LZ} : 0.9776

- $C_{\text{dev-R}}$: 0.7758
- $C_{\text{DFG-E}}$: 0.8709
- $C_{\text{DFG-cnc}}$: 0.8683
- $C_{\text{DFG-mdeg}}$: 0.9679
- $C_{\text{DFG-dens}}$: 0.7419
- $C_{\text{DFG-struct}}$: 0.7448
- $C_{\text{DFG-CN}}$: 0.8446
- $C_{\text{DFG-sep-#}}$: 0.94
- $C_{\text{DFG-sep-r}}$: 0.6112
- $C_{\text{DFG-cyc}}$: 0.9349
- $C_{\text{avg-dist}}$: 0.9221
- $C_{\text{DA-min}}$: 0.9664
- overall MSA: 0.882187383709089

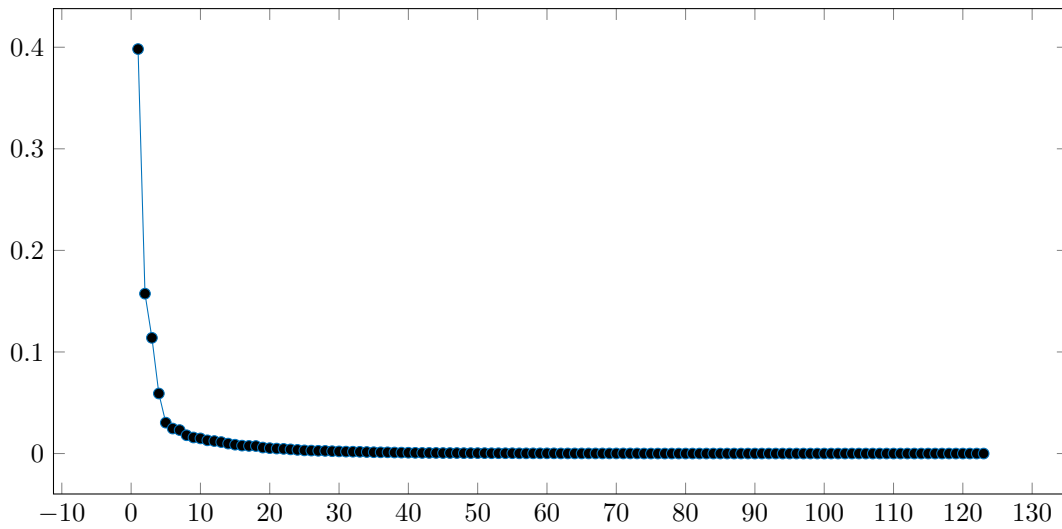
During the analysis, the MSA values of the following complexity measures were too low:

- $C_{\text{A-start-IQR}}$: 0.2879
- $C_{\text{A-end-IQR}}$: 0.324
- C_{struct} : 0.281
- C_{affinity} : 0.4221
- $C_{\text{DFG-seq-r}}$: 0.2501

These complexity measures were left out of the analysis to produce dependable results.

Chosen Number of Factors

The Scree test resulted in the following Eigenvalue plot:



You choose to set the number of expected factors to 10.